

FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 – 1.3 CELL BIOLOGY: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving Question: Why do the cells in my body, and in the sukuma wiki on my plate, look so different from each other, and how does each cell's structure allow it to do its specific job? Remember the Kisumu blood smear: red blood cells with no nucleus next to white blood cells with large nuclei, and the sukuma wiki leaf cell that is rectangular, has chloroplasts and a large central vacuole. Answer every section in the space under each prompt. Write in full sentences and use the scientific words you learned in Lessons 1 to 8. Use evidence from the images, practicals and activities we did. Where a prompt asks for a Claim, Evidence and Reasoning (CER) answer, state your claim, back it with evidence, and then explain why the evidence supports your claim. Work on your own.

Part 1: What do all the cells share?

The red blood cell, the white blood cell and the sukuma wiki leaf cell all look very different, yet we call all of them cells. Bacteria in fermenting uji are also called cells. (a) Name the structures or features that ALL cells share. (b) Use a Claim, Evidence and Reasoning (CER) paragraph to explain why bacteria count as true cells even though they have no membrane-bound nucleus. (c) Why do you think these shared features are essential for life?

Write your answer here:

Part 2: Plant cell or animal cell?

Use what you saw in the onion and cheek cell practical and in the sukuma wiki leaf cell. (a) Describe three structures found in the sukuma wiki leaf cell but NOT in the human blood cells, and say what each one does. (b) Explain why the sukuma wiki cell looks rectangular but a cheek cell looks irregular and rounder. (c) The onion epidermal cells we viewed were not green, although onion is a plant. Suggest a reason, using what you learned about the organelle responsible for the green colour.

Write your answer here:

Part 3: Structure and function in specialised cells

Look back at the blood smear from the Kisumu County Hospital. (a) The red blood cells have no nucleus and are biconcave discs. Explain how this structure is linked to the job of the red blood cell. (b) The white blood cells have large nuclei. Using what you learned about the nucleus, explain what the nucleus allows a cell to do that a mature red blood cell cannot do. (c) Choose ONE other specialised cell (guard cell, root hair cell, muscle cell, nerve cell or sperm cell). Describe its special structure and explain how it helps the cell do its job, with a Kenyan example from the sukuma wiki plant, a bean seedling or the human body.

Write your answer here:

Part 4: When structure changes, function changes

In sickle cell anaemia, some red blood cells have a crescent shape instead of the normal shape. Explain how the change in structure affects the function of the cell and what this means for a person's body. Then explain, using this example, why structure and function are linked in cells.

Write your answer here:

Part 5: Answer the driving question

Write a full explanation to answer the driving question: Why do the cells in my body, and in the sukuma wiki on my plate, look so different from each other, and how does each cell's structure allow it to do its specific job? Your answer must include: (1) the structures all cells share, (2) the main differences between plant and animal cells, (3) the idea of specialisation, explaining what is added, removed or modified, and (4) one Kenyan example that shows structure matching function.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Shared cell features and cell types	Names all four universal features (membrane, cytoplasm, DNA, ribosomes), correctly distinguishes prokaryotes from eukaryotes and uses a full CER with evidence and reasoning.	Names most universal features and states that bacteria are cells, but evidence or reasoning is incomplete.	Names few shared features or confuses what makes something a cell.
Plant versus animal cell structures	Correctly names cell wall, chloroplasts and central vacuole, explains the function of each and links them to the shape and colour of the sukuma wiki cell.	Names the plant structures with some functions, with limited links to the observed cells.	Lists structures without functions or mixes up plant and animal features.
Structure-function reasoning in specialised cells	Explains clearly how the structure of the red blood cell, white blood cell and another specialised cell suits its job, including the trade-off of losing the	Links structure to function correctly for most examples but with limited detail.	States structures or functions separately without linking them.

	nucleus.		
Use of evidence and Kenyan context	Draws consistently on the blood smear, sukuma wiki, practicals and sickle cell example, accurately and in the same way as taught.	Uses some evidence and examples from the lessons.	Uses little or no evidence from the lessons, or inaccurate examples.
Communication of the final explanation	Answer to the driving question is organised, complete (all four required parts), and uses scientific vocabulary accurately.	Answer covers most required parts with mostly correct vocabulary.	Answer is incomplete or unclear, with frequent vocabulary errors.

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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